Progressive exercises

E1 · Practice. Find all the antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Practice. Find the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ that satisfies $F(0) = 0$.

E3 · Practice. Find the solution on $\mathbb{R}$ of the differential equation $y' = 3y$ that satisfies $y(1) = 2$.

E4 · Practice. Find the solution on $\mathbb{R}$ of the differential equation $y' = -y + 4$ that satisfies $y(0) = 7$.

E5 · Practice. Consider the differential equation $(E) : y' = y - x$ on $\mathbb{R}$. Check that $g(x) = x + 1$ is a solution of it, then find the solution of (E) that satisfies $y(0) = 0$.

Textbook or past-paper work

Barbazo, p. 228, exercises 25 and 30. In S55: exercise 30, to bring out a constant solution. At home: exercise 25 in full, distinguishing a condition on f from a condition on f'.